

ADR 001 - Rework Health message (9.3.2)

Status	PROPOSED
Date	Jan 14, 2026
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Consulted	
Informed	

Context and Problem Statement

Currently, a health message contains only the NAMUR NE107 code and a health score. Any further description of a health message, like a diagnostic code needs to be provided as an event message. Event messages, on the other hand, cannot be retrieved via GET. This means that if an event message is missed, there is no way to access this information again.

Another issue with the current structure is that an asset can only have a single health status. In practice, however, multiple health statuses can overlap. For example, a consumable might be running low, triggering a "Maintenance Required" status, while at the same time, a measurement value outside the specification results in an "Off Spec" status. However, these issues can be superimposed by a more severe, currently active failure.

Statuses with lower severity, such as "Check Function" or "Maintenance Required," are often ignored by operators until they worsen and ultimately lead to a failure.

Decision Drivers

- Additional health information should be available at the health directly and not at the health event
- More than one status should be possible per asset (Printer can run out of toner and paper-jam at the same time ⇒ two NE107 status)

Decision Outcome/ Chosen Option

The health messages shall be changed to the following structure:

Health :

Value Range: `<DeviceHealthEnumeration>`

Type: DeviceHealthEnumeration has TypeOf enumeration. It is defined for OPC UA-JSON as `<name>_<value>` (OPC UA [Part 6-5.4.4](#)).

Access: Read only

Requirement: Mandatory

Health indicates the most sever issue/status as defined by the NAMUR recommendation NE107 and its type is `DeviceHealthEnumeration` (OPC UA [Part 100-4.5.4](#)).

The enumeration defines the asset condition, which is described in OPC UA [Part 100-4.5.4-Table 22](#).

The best practice is to combine the five states `NORMAL_0` , `FAILURE_1` , `CHECK_FUNCTION_2` , `OFF_SPEC_3` and `MAINTENANCE_REQUIRED_4` with the symbol/color definitions made by NAMUR NE107 ([table 12](#)).

HealthScore :

Value Range: `<Byte>`

Type: Byte in a range of 0..100 %

Access: Read only

Requirement: Optional

HealthScore reflects a meter to indicate the current health level as a result of sub-optimal process and/or environmental conditions in the range from 0 to 100 %.

Details[] :

Value Range: `<array of DetailObject>`

Type: `ComponentsObject`

Access: Read only

Requirement: Optional

The **Details** list contains all subordinate health statuses of an asset or service.

The **DetailObject** is structured as following:

Health :

Value Range: `<DeviceHealthEnumeration>`

Type: DeviceHealthEnumeration has TypeOf enumeration. It is defined for OPC UA-JSON as `<name>_<value>` (OPC UA [Part 6-5.4.4](#)).

Access: Read only

Requirement: Mandatory

Health indicates status as defined by the NAMUR recommendation NE107 and its type is `DeviceHealthEnumeration` (OPC UA [Part 100-4.5.4](#)).

The enumeration defines the asset condition, which is described in OPC UA [Part 100-4.5.4-Table 22](#).

The best practice is to combine the five states `NORMAL_0` , `FAILURE_1` , `CHECK_FUNCTION_2` , `OFF_SPEC_3` and `MAINTENANCE_REQUIRED_4` with the symbol/color definitions made by NAMUR NE107 ([table 12](#)).

DiagnosticCode :

Value Range: `<event condition>`

Type: String

Access: Read only

Requirement: Optional

Manufacturer specific detail information about the the event - e.g., F-238.

Location :

Value Range: `<location within the asset/service>`

Type: String

Access: Read only

Requirement: Optional

Actual location of the raised health event - e.g., if an asset has multiple sensor units build like temperature.

Description :

Value Range: <Description>

Type: String

Access: Read only

Requirement: Optional

Human readable **Description** related to the NE107 status code used in **Health**.

Optional Outcome

By adding the health event information to the health message, the event type NE107 becomes obsolete as change in the health message requires the service to send the resource health anyway.

Consequences/ Pros and Cons

- ✔ With the new solution, both additional descriptions for a health message can be provided, and the practical case of multiple concurrent health statuses can be represented.
- ✔ Non-breaking change as only optional elements are appended
- ✔ NE107-Event becomes obsolete
- ✖ Guideline must be adjusted
- ✖ SDK / Toolkits / Service implementations must be adjusted